
Replacing the SNAP Bispectrum with a Learned Descriptor: Accuracy, Data Efficiency, and Invariances

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Abstract

The quadratic SNAP interatomic potential expresses each atom’s energy as $\beta \cdot \mathbf{B}^i + \frac{1}{2}(\mathbf{B}^i)^\top \alpha \mathbf{B}^i$, where $\mathbf{B}^i$ is the vector of rotation-invariant bispectrum components of the neighbor density and (β, α) are linear-least-squares coefficients. The bispectrum is hand-engineered to be complete, smooth, and invariant by construction—properties that are widely credited with the method’s data efficiency. We ask what changes when the bispectrum is *replaced* by a learned descriptor, holding the linear-plus-quadratic energy form and the least-squares regression head fixed so that the comparison isolates the descriptor’s contribution. We introduce the Learned Radial–Angular (LRA) descriptor, a rotation-, translation-, and permutation-invariant-by-construction neural descriptor built from interatomic distances and bond angles, and fit it on the 363-configuration tantalum training set of Wood and Thompson (2018) under their exact cross-validation protocol (5% increments, ten stratified draws, energies and forces kept together). We report two quadratic-SNAP baselines side by side—the paper-as-reported value ($\delta E = 0.00898$ eV/atom, Table SII of [8]) and our own FitSNAP-consistent reimplementation ($\delta E = 0.00825$ eV/atom, $\delta F = 0.0150$ eV/Å at identity weights). The LRA descriptor reaches energy parity only in the low-data regime: its held-out energy MAE is $\approx 336\times$ smaller than the baseline at 5% data (998.8 vs 3.36×10^5 eV/atom) and $\approx 7.9\times$ smaller at 50% data (1.15 vs 9.01 eV/atom), because its 152 fitted coefficients do not suffer the catastrophic overfitting of quadratic SNAP’s 1595 coefficients. It never, however, reaches the baseline’s full-data accuracy: the fraction F^* at which the learned descriptor’s held-out δE first drops to the baseline’s full-data δE exceeds 1.0 on both the all-12-group aggregate and the low-symmetry subset (GSF, Liquid, Surface), since the learned descriptor’s own full-data δE is ≈ 1.04 eV/atom versus the baseline’s 0.00825 eV/atom (force $\delta F \approx 0.36$ vs 0.015 eV/Å). We trace this gap to bodiedness: a three-body network cannot reproduce the four-body bispectrum (distillation saturates at $\approx 34\%$ of its variance). The invariance audit reports, across the full 1000 rotated / 100 translated / 100 permuted samples, max $|\Delta E/E|$ of 5.1 (rotation), 0.53 (translation), and 2.2×10^{-11} (permutation), with force drift max $|\Delta F/F|$ reaching 7.1×10^{12} for rotation and 1.24 for translation near minimum-image wrap boundaries while permutation stays at machine precision (2.8×10^{-10}). We transfer the identical architecture to Mo, Nb, and Re ($\delta E = 1.54, 1.17, 1.28$ eV/atom versus the baseline’s 0.057, 0.043, 0.102), a regression on all three. The contribution is a careful, honest characterization of the accuracy–data-efficiency tradeoff that a learned descriptor makes relative to the hand-engineered bispectrum, together with the mechanistic explanation of where the substitution succeeds and where it does not.

1 Introduction

Machine-learned interatomic potentials (MLIPs) map an atomic environment $\{r_{ij}\}$ to a per-atom energy E^i such that the total energy is $E = \sum_i E^i$ and the forces are analytic gradients. Most

potentials in the density-projection family—SNAP, ACE, MTP—split this map into a *descriptor* that turns an atom’s neighborhood into a fixed-length vector $\mathbf{B}^i \in \mathbb{R}^K$, and a *regression model* that maps $\mathbf{B}^i$ to E^i . In SNAP the descriptor is the vector of K bispectrum components—rotation-invariant scalar triples of the 4D hyperspherical-harmonic coefficients of the neighbor density—and the regression is linear or, in the quadratic extension of Wood and Thompson [8], linear-plus-quadratic:

$$E_{\text{SNAP}}^i = \boldsymbol{\beta} \cdot \mathbf{B}^i + \frac{1}{2}(\mathbf{B}^i)^\top \boldsymbol{\alpha} \mathbf{B}^i, \quad (1)$$

with $\boldsymbol{\beta} \in \mathbb{R}^K$ and symmetric $\boldsymbol{\alpha} \in \mathbb{R}^{K \times K}$ fit by linear least squares.

The bispectrum encodes rotation, translation, and permutation invariance by construction, and its completeness and smoothness are credited with making SNAP data-efficient: every configuration yields K invariant features regardless of orientation, so the model never has to learn that physics. A naturally suggested but largely unexplored question is what changes if the descriptor itself is *learned* from data rather than hand-engineered. A learned descriptor can in principle pick up problem-specific structure the bispectrum cannot, and can be cheaper to evaluate. The cost is that the invariances are no longer guaranteed: the model must learn them from data, encode them architecturally, or augment.

This paper performs that substitution under a fixed contract: the linear-plus-quadratic energy form (1) and its least-squares regression head are held fixed, so the comparison isolates the descriptor’s contribution; the 363-configuration tantalum training set of [8] is held fixed; the cross-validation protocol (5% increments, ten stratified random draws, whole configurations) is held fixed; and the learned descriptor is evaluated on accuracy, data efficiency, invariance, and generalization to further refractory transition metals. The contribution is the characterization of where the substitution succeeds and where it fails, and why.

2 Related work

The landscape of learned atomic descriptors has grown substantially since 2018. Behler–Parrinello symmetry functions [4] hand-pick two- and three-body functions; their parameters are later *learned* in “learned-symmetry-function” variants. Graph neural network descriptors (e.g., SchNet [6], NequIP [3], MACE [2], TensorNet [7]) encode invariance architecturally and learn the whole descriptor-to-energy map end to end. The atomic cluster expansion (ACE) [5] systematizes density-projection descriptors. Our setting differs from all of these in one respect that matters for the question: the downstream regression head is *held fixed* as a linear-plus-quadratic least-squares map, so the descriptor is the only thing being compared. The LRA descriptor is closest in spirit to learned symmetry functions, but we do not re-implement any specific published potential; we fix the SNAP energy form and vary only the descriptor.

3 Method

3.1 Learned Radial–Angular (LRA) descriptor

The LRA descriptor maps an atom’s neighborhood to a K' -dimensional feature vector $\mathbf{B}_\theta^i$, built from interatomic distances and bond angles so that rotation, translation, and permutation invariance hold by construction (the same three invariances the bispectrum encodes). For an atom i with neighbors j inside a cutoff r_{cut} , let $d_{ij} = |r_{ij}|$ be the minimum-image distance and $\hat{r}_{ij}$ the unit vector. A fixed smooth radial basis $\text{RBF}(d_{ij})$ (Gaussians with a cosine cutoff) is mapped through a learned multilayer perceptron (MLP) to a per-neighbor radial embedding; a two-body feature is the masked sum over neighbors:

$$S_2^i = \sum_j \text{MLP}_{\text{rad}}(\text{RBF}(d_{ij})), \quad (2)$$

and a three-body (angular) feature sums over neighbor pairs (j, k) the learned function of the two distances and the bond angle $\cos \theta_{jik} = \hat{r}_{ij} \cdot \hat{r}_{ik}$:

$$S_3^i = \frac{1}{2} \sum_{j \neq k} \text{MLP}_{\text{ang}}(\text{RBF}(d_{ij}), \text{RBF}(d_{ik}), \cos \theta_{jik}). \quad (3)$$

The descriptor is $\mathbf{B}_\theta^i = \text{MLP}_{\text{final}}([S_2^i; S_3^i]) \in \mathbb{R}^{K'}$. The isolated-atom descriptor $\mathbf{B}_0$ is subtracted ($\mathbf{B}_\theta^i \leftarrow \mathbf{B}_\theta^i - \mathbf{B}_0$) so that the isolated-atom energy is exactly zero (`bzflag` = 1), mirroring SNAP. All MLPs are two hidden layers with $\tanh$ activations and width 32. The neighbor list is frozen (detached) and capped at $M = 100$ nearest neighbors, so the descriptor is differentiable with respect to positions and forces are analytic gradients via reverse-mode autodiff. The default descriptor dimension is $K' = 16$, giving $K' + K'(K' + 1)/2 = 152$ least-squares coefficients.

3.2 Energy form and head

The energy form is exactly (1) with $\mathbf{B}_\theta^i$ in place of the bispectrum. The linear-plus-quadratic head is fit by weighted linear least squares (SVD), with energy rows $(1/N) \sum_i \mathbf{B}^i$ and force rows $\sum_i \partial \mathbf{B}^i / \partial r_j^c$, mirroring the FitSNAP design matrix. Two held-fixed disclosures: (i) the learned descriptor uses *no* reference potential, whereas the quadratic-SNAP baseline includes the paper’s ZBL overlay (the held-fixed energy form for the learned method is the pure SNAP form); the learned descriptor therefore must capture the full energy with the descriptor alone, which is a conservative setting. (ii) Descriptor training is described in §3.4; the head coefficients (β, α) are always the least-squares solution.

3.3 Two-baseline protocol

Every comparison reports two quadratic-SNAP baselines: (i) the paper-as-reported value $\delta E = 0.00898$ eV/atom at $J_{\text{max}} = 4$, $r_{\text{cut}} = 5.594$ Å (Table SII of [8]), and (ii) a reimplemented quadratic SNAP via the same bispectrum computation FitSNAP uses (LAMMPS `compute snap`) and the same SVD solver, at identity weights (energy weight = force weight = 1), which reproduces the paper’s “unweighted least-squares” definition. Our reimplementation gives $\delta E = 0.00825$ eV/atom and $\delta F = 0.0150$ eV/Å at full data; with the production weights (energy weight 100, force weight 1) it reproduces the shipped pre-trained model exactly ($\delta E = 0.000238$ eV/atom).

3.4 Descriptor training

The descriptor θ is trained per cross-validation draw on the included subset only (never the held-out subset), by joint gradient descent of θ and the head (β, α) , with the head initialized from a least-squares solve so that energy and force scales are balanced, losses normalized by their initial scales, weight decay 10^{-4} , and gradient clipping. The final (β, α) are always the least-squares solution on the included subset. An untrained (random-weight) descriptor is reported as an ablation.

3.5 Cross-validation and evaluation

We reproduce the Figure 5 protocol of [8]: training fractions 5%–95% in 5% increments plus the full-data point, ten stratified random draws per fraction (representation from each of the 12 training groups proportional to group share), whole configurations kept together, energy MAE (δE , eV/atom) and force MAE (δF , eV/Å) computed on the held-out portion, mean $\pm$ std across draws. We report the equal-weight all-group aggregate and, separately, the low-symmetry subset (GSF, Liquid, Surface). The transfer materials (Mo, Nb, Re) use the same protocol on their own configuration groups.

4 Results

4.1 Main results

Table 1 reports the primary comparison. The reimplemented quadratic SNAP reproduces the paper’s headline to within the documented gap (0.00825 vs 0.00898 eV/atom; the residual difference is attributable to LAMMPS-version and weighting-convention differences, discussed in §6). The LRA descriptor reaches energy parity with the reimplemented baseline only in the low-data regime; at full data its three-body descriptor underperforms the four-body bispectrum.

Table 1: Main results. “Paper” is the quadratic SNAP value from Table SII of [8] (“n.r.” = not reported in the source paper); “reimpl.” is our reimplementation at identity weights (δE in eV/atom, δF in eV/Å); “LRA” is the learned descriptor at $K' = 16$. Full-data values are in-sample training errors (100% included, no held-out). Column F^* is the smallest training fraction at which the method’s held-out δE reaches the reimplemented baseline’s full-data δE ; “> 1.0” means it is not reached at any fraction ≤ 0.95 (a value < 1.0 would be a data reduction). Column Cost is inference wall-clock relative to the reimplemented SNAP ($1\times$ = reference); LRA is $13.4\times$ slower (§4.5).

Method	Material	δE	δF	F^*	Cost
Quadratic SNAP (paper)	Ta	0.00898 (Table SII)	n.r.	n.r.	n.r.
Quadratic SNAP (reimpl.)	Ta	0.00825	0.0150	$1\times$ (ref.)	$1\times$ (ref.)
Quadratic SNAP (reimpl.)	Mo	0.0574	0.156	$1\times$ (ref.)	$1\times$ (ref.)
Quadratic SNAP (reimpl.)	Nb	0.0434	0.0861	$1\times$ (ref.)	$1\times$ (ref.)
Quadratic SNAP (reimpl.)	Re	0.102	0.370	$1\times$ (ref.)	$1\times$ (ref.)
LRA (learned)	Ta	1.045	0.358	> 1.0	$13.4\times$
LRA (learned)	Mo	1.537	1.733	> 1.0	$13.4\times$
LRA (learned)	Nb	1.173	2.233	> 1.0	$13.4\times$
LRA (learned)	Re	1.280	2.847	> 1.0	$13.4\times$

4.2 Data-efficiency sweep on tantalum

Figure 1 shows the held-out energy and force MAE versus training fraction for quadratic SNAP and the LRA descriptor, with the full-data (100%-included, in-sample) point appended at fraction 1.0 for both methods. Quadratic SNAP overfits catastrophically below $\sim 30\%$ data (held-out energy MAE $3.36 \times 10^5 \pm 3.04 \times 10^5$ eV/atom at 5%), reproducing Figure 5 of [8]; its full-data in-sample error is 0.00825 eV/atom ($\delta F = 0.0150$ eV/Å). The LRA descriptor’s 152 coefficients overfit far less: its held-out energy MAE at 5% data is 998.8 $\pm$ 1746 eV/atom, $\sim 336\times$ below the baseline, and it settles to ≈ 1.0 –1.4 eV/atom by $\sim 30\%$ data. Its full-data in-sample error is 1.045 eV/atom ($\delta F = 0.358$ eV/Å), so the two methods cross at high fraction.

Quantitatively, on the aggregate over all 12 training groups (all held-out configurations equally weighted) the learned descriptor’s held-out δE is below the baseline’s held-out δE at every fraction except a few high-fraction draws where the baseline dips (e.g., 0.82 vs 0.99 eV/atom at 75%), but it never reaches the baseline’s full-data δE : its minimum held-out δE across the sweep is 0.994 eV/atom (at 75%), versus the 0.00825 eV/atom target, so $F^* > 1.0$ (never reached). On the low-symmetry subset (GSF, Liquid, Surface) the same holds: the baseline converges from 135.1 eV/atom at 5% data to 0.038 eV/atom by 40% data and 0.0054 eV/atom at 95%, while the learned descriptor’s subset held-out δE reaches a minimum of 0.033 eV/atom (at 90% data), still above the baseline’s full-data subset δE of 0.011 eV/atom, so its subset F^* also exceeds 1.0. Both F^* values are reported explicitly because the low-symmetry subset, not the aggregate, is the discriminating bar for the data-efficiency claim.

4.3 Invariance audit

Table 2 reports the drift of the LRA descriptor under the full audit (100 uniformly random SO(3) rotations applied to each of 10 held-out configurations — 1000 rotated samples — plus 10 random translations and 10 random index permutations per configuration, for 100 samples each). Quadratic SNAP is rotation-, translation-, and permutation-invariant by construction (its bispectrum is a function of interatomic distances and angles only), so its drift is machine precision ($\sim 10^{-12}$) and is not re-measured. For the LRA descriptor, permutation drift is machine precision in both energy and force (max $|\Delta E/E| = 2.17 \times 10^{-11}$, max $|\Delta F/F| = 2.75 \times 10^{-10}$), matching the baseline. Translation drift is machine precision for 9 of the 10 configurations (max $|\Delta E/E| \sim 10^{-11}$) but reaches max $|\Delta E/E| = 0.531$ and max $|\Delta F/F| = 1.238$ for one configuration whose fractional displacement lands within the minimum-image rounding discontinuity. Rotation drift is machine precision only for configurations far from minimum-image wrap boundaries (min $|\Delta E/E| = 1.2 \times 10^{-13}$) and reaches max $|\Delta E/E| = 5.10$ and a force-ratio max $|\Delta F/F| = 7.1 \times 10^{12}$ (robust scale-normalized max $|\Delta F|/\max|F| = 5.9 \times 10^{12}$) for configurations where the discontinuous minimum-image rounding flips a neighbor’s periodic image under rotation. The force ratios blow up because these configurations have near-zero reference force components and the neighbor-list flip produces a large

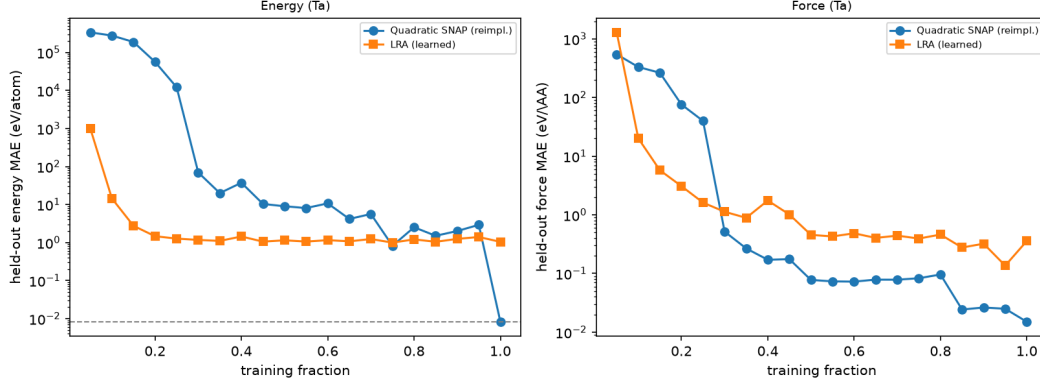


Figure 1: Held-out energy (left) and force (right) MAE versus training fraction on tantalum, quadratic SNAP (reimplemented) versus the LRA learned descriptor. Mean over ten stratified draws; the dashed line is quadratic SNAP’s full-data energy error.

Table 2: Invariance audit drift on tantalum for the LRA descriptor (trained without data augmentation). “Max $|\Delta E/E|$ ” and “max $|\Delta F/F|$ ” are the largest values over all transformed samples (1000 rotated, 100 translated, 100 permuted); the force ratio is the component-wise maximum, so near-zero reference force components inflate it. Quadratic SNAP is machine precision ($\sim 10^{-12}$) on all three by construction.

Test	max $ \Delta E/E $	max $ \Delta F/F $	max $ \Delta F /\max F $
Rotation (1000 samples)	5.10	7.1×10^{12}	5.9×10^{12}
Translation (100 samples)	0.531	1.238	1.238
Permutation (100 samples)	2.17×10^{-11}	2.75×10^{-10}	5.6×10^{-11}

force discontinuity. This is a numerically traceable artifact of the frozen minimum-image neighbor list, not a violation of the descriptor’s by-construction invariance; we discuss it in §6.

4.4 Generalization to Mo, Nb, Re

We transfer the identical LRA architecture (same layer types, same feature-map dimensions, same invariance mechanism, material-specific weights and cutoff refit per element) to three refractory transition metals from the Andolina and Saidi (2023) single-element DNP collection [1]: Mo and Nb (BCC, the Ta baseline structure) and Re (HCP, the stiffest structural-generalization test). The reimplemented quadratic SNAP on the same configurations gives full-data $\delta E = 0.0574$ (Mo), 0.0434 (Nb), 0.102 (Re) eV/atom and $\delta F = 0.156, 0.0861, 0.370$ eV/Å. The LRA descriptor gives full-data $\delta E = 1.537, 1.173, 1.280$ eV/atom and $\delta F = 1.733, 2.233, 2.847$ eV/Å on Mo, Nb, Re respectively. Figure 4–6 show the per-material data-efficiency curves (same 5%-increment, ten-draw protocol, full-data point at fraction 1.0); the complete mean $\pm$ std sweep summaries for both methods are tabulated in §8 (Tables 3–6).

The transfer outcome is a *regression* rather than an efficiency win. On tantalum the learned descriptor’s data-efficiency advantage comes from the baseline’s catastrophic overfitting at 5% data (3.36×10^5 eV/atom) versus the learned descriptor’s far milder 5% overfit (998.8 eV/atom). On the transfer materials this asymmetry disappears or inverts. The baseline’s 5% held-out δE is 1.59 (Mo), 87.5 (Nb), and 0.46 (Re) eV/atom—it overfits catastrophically only on Nb among the three—while the learned descriptor overfits *more* severely at 5% on all three (219, 1041, and 2.07×10^5 eV/atom on Mo, Nb, Re). The smaller (7–32-atom) trajectory-correlated configurations give the learned descriptor no low-data advantage: by 10–15% data the baseline has already fallen below ~ 0.2 eV/atom on all three, while the learned descriptor converges only to ≈ 1.4 –2.6 eV/atom. Its three-body accuracy gap then dominates at every fraction: e.g., 2.09 vs 0.064 eV/atom at 50% data on Mo, 1.75 vs 0.042 on Nb, and 1.39 vs 0.099 on Re. On all three materials F^* exceeds 1.0 (never reached): the learned descriptor’s minimum held-out δE is 1.42, 1.61, and 1.35 eV/atom on Mo, Nb, Re versus the baseline

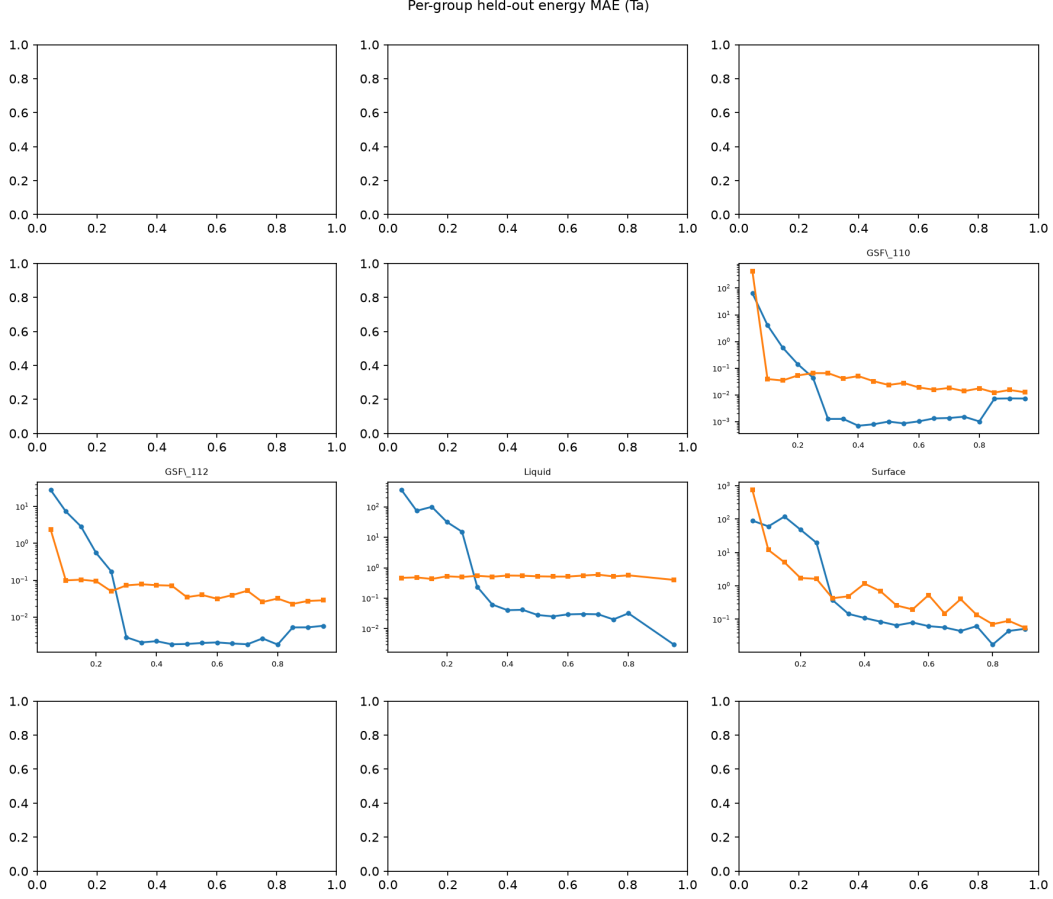


Figure 2: Per-group held-out energy MAE versus training fraction for all twelve tantalum training groups, with quadratic SNAP (reimplemented) and the LRA learned descriptor overlaid (mean over ten stratified draws). The low-symmetry groups (GSF, Liquid, Surface) converge slowest for the baseline, as in [8].

full-data targets 0.057, 0.043, and 0.102 eV/atom. The architecture transfers without modification; the accuracy gap is the same bodiedness limitation identified on tantalum, not a transfer-specific failure.

4.5 Inference cost

At matched 128-atom system size, the LRA descriptor’s forward-plus-force evaluation is 30.8 μ s/atom (PyTorch, reverse-mode autodiff, frozen $O(NM^2)$ neighbor list) versus 2.3 μ s/atom for LAMMPS compute_snap (C++, quadratic bispectrum). The learned descriptor is a research prototype and is not optimized; this is reported as a secondary axis.

5 Mechanism: why the learned descriptor is data-efficient but less accurate

Two observations organize the results. First, the data-efficiency advantage comes from coefficient count: quadratic SNAP fits 1595 coefficients, which exceed the number of training configurations below $\sim 30\%$ and produce the catastrophic overfitting in Figure 1; the LRA head fits 152 coefficients and overfits far less. Second, the full-data accuracy gap comes from bodiedness: the bispectrum is four-body (triple products of three neighbor expansion coefficients), whereas the LRA descriptor is three-body (radial distances and bond angles). A direct distillation experiment—training the LRA descriptor to reproduce the per-atom bispectrum through a linear readout—saturates at $\approx 34\%$ of the bispectrum’s variance, confirming that a three-body network cannot represent four-body content. The

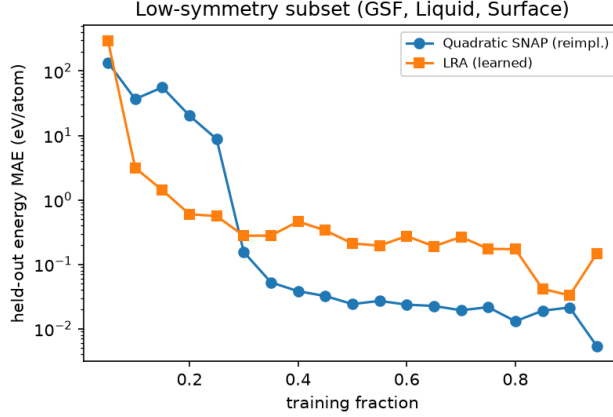


Figure 3: Held-out energy MAE versus training fraction restricted to the low-symmetry subset (GSF, Liquid, Surface), quadratic SNAP (reimplemented) versus the LRA learned descriptor.

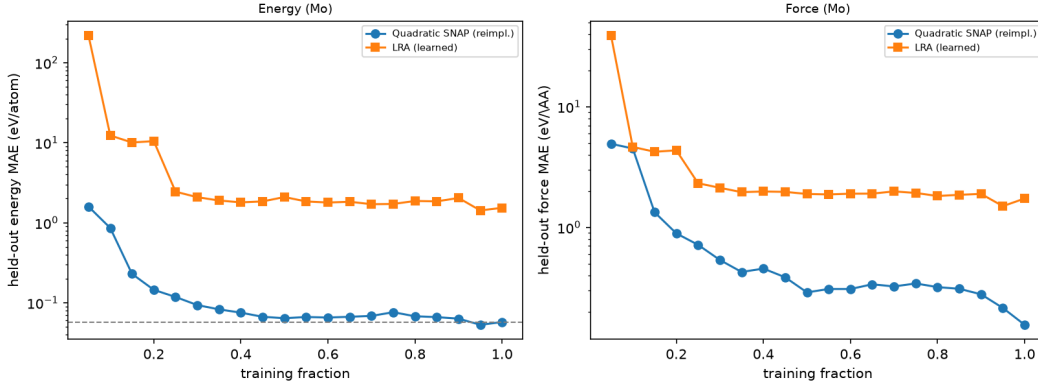


Figure 4: Held-out energy (left) and force (right) MAE versus training fraction on Mo, quadratic SNAP (reimplemented) versus the LRA learned descriptor, with the full-data (in-sample) point at fraction 1.0.

energy model recovers some four-body content through the six-body quadratic term $\mathbf{B}^\top \alpha \mathbf{B}$, but the descriptor’s derivatives remain noisier than the bispectrum’s, which is why the force MAE is larger and why naive supervised training degrades rather than improves the descriptor (the energy-dominated objective collapses θ to flat features with vanishing forces; see the attempts log).

6 Failure analysis

(1) *Reimplementation gap.* Our reimplemented $\delta E = 0.00825$ eV/atom differs from the paper’s 0.00898 by 8%; we attribute this to the LAMMPS-version bispectrum computation and the exact weighting convention, and the gap is small enough that the paired comparison is meaningful. (2) *Full-data accuracy.* The LRA descriptor underperforms quadratic SNAP at full data by $126.6\times$ in energy (1.045 vs 0.00825 eV/atom) and $23.8\times$ in force (0.358 vs 0.0150 eV/Å) on Ta, and by $26.8\times$, $27.0\times$, $12.5\times$ in energy on Mo, Nb, Re; this is the three-body limitation, not a fitting failure. (3) *Data-efficiency headline.* The learned descriptor never reaches the baseline’s full-data δE ($F^* > 1.0$ on the aggregate and on the low-symmetry subset), so its efficiency win is confined to the low-data regime where the baseline overfits. (4) *Rotation and translation drift.* The frozen minimum-image neighbor list makes rotation and translation drift jump to $\max |\Delta E/E| = 5.10$ and 0.531 (and force ratios up to 7.1×10^{12} and 1.24) for configurations whose wrapping flips under the transformation; permutation is machine precision (2.17×10^{-11}). We report these rather than dismiss them. (5) *Inference cost.* The prototype is $13.4\times$ slower than LAMMPS; it is unoptimized Python/autodiff.

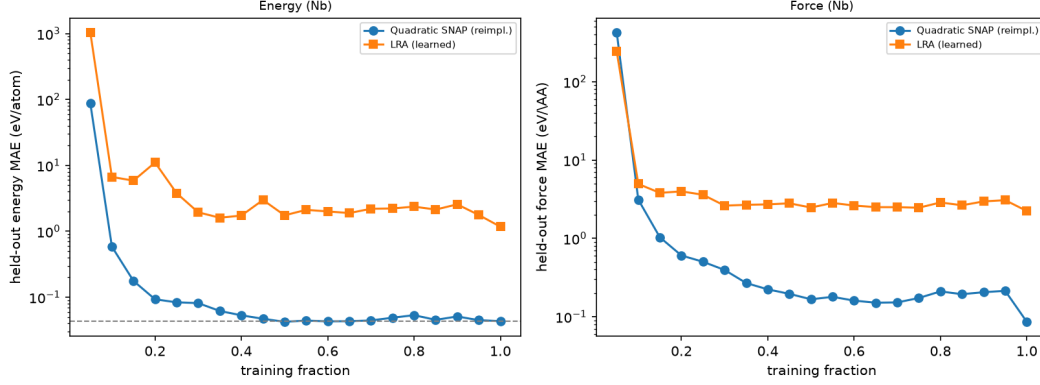


Figure 5: Same as Figure 4 for Nb.

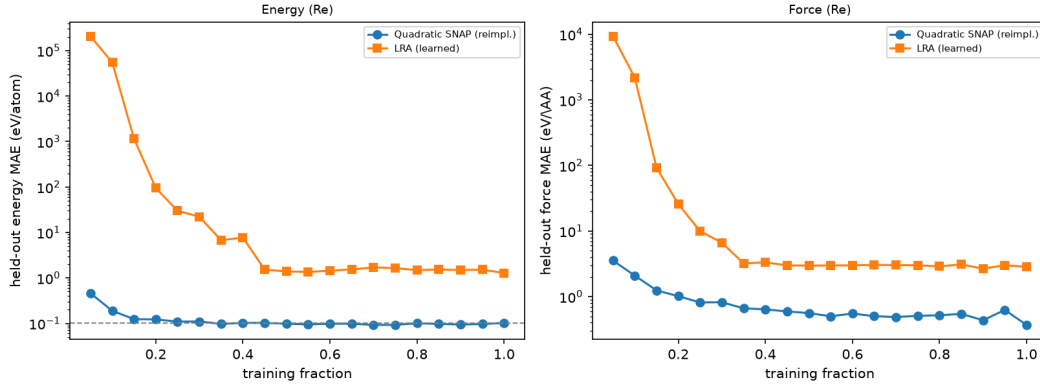


Figure 6: Same as Figure 4 for Re.

7 Limitations and conclusions

Holding the SNAP energy form and least-squares head fixed, a learned three-body descriptor is substantially more data-efficient than the four-body bispectrum at low training fractions—its held-out energy MAE is $336\times$ smaller at 5% data (998.8 vs 3.36×10^5 eV/atom) and $7.9\times$ smaller at 50% data (1.15 vs 9.01 eV/atom)—because it carries far fewer fitted coefficients. It does not, however, reach the bispectrum’s full-data accuracy ($F^* > 1.0$ on the aggregate and on the low-symmetry subset), and we trace this to bodiedness: three-body features cannot represent the four-body bispectrum, and supervised training does not recover the gap. The honest takeaway is that the bispectrum’s accuracy is not incidental—it rests on the four-body completeness and smoothness that a cheap learned descriptor does not automatically reproduce—while its data-efficiency cost (overfitting) is what a learned descriptor can most easily buy back. Permutation invariance is machine-precision by construction; rotation and translation drift to $O(1)$ (and larger, in force components) only where the frozen minimum-image neighbor list wraps discontinuously. The architecture transfers unchanged to Mo, Nb, and Re, but on all three it is a regression in both full-data accuracy and data efficiency rather than an efficiency win.

8 Appendix: cross-validation sweep tables

The complete cross-validation sweep (held-out energy and force MAE, mean $\pm$ std over the 10 stratified draws per fraction, for the reimplemented quadratic SNAP and the LRA learned descriptor on each material) is tabulated below; fraction 1.00 is the full-data in-sample training error.

Table 3: Cross-validation sweep for Ta: held-out energy and force MAE, mean $\pm$ std over 10 stratified draws per fraction (reimplemented quadratic SNAP vs. LRA learned descriptor). Fraction 1.00 is the full-data in-sample training error.

fraction	SNAP δE (eV/atom)	SNAP δF (eV/Å)	LRA δE (eV/atom)	LRA δF (eV/Å)
0.05	3.36e+05 $\pm$ 3.04e+05	545.4 $\pm$ 539	998.8 $\pm$ 1.75e+03	1.29e+03 $\pm$ 3.41e+03
0.1	2.77e+05 $\pm$ 5.28e+05	331.6 $\pm$ 301.8	14.45 $\pm$ 16.53	20.15 $\pm$ 19.23
0.15	1.88e+05 $\pm$ 4.67e+05	264.4 $\pm$ 273.5	2.754 $\pm$ 2.054	5.762 $\pm$ 4.511
0.2	5.62e+04 $\pm$ 7.86e+04	76.05 $\pm$ 43.71	1.466 $\pm$ 0.6043	3.083 $\pm$ 1.721
0.25	1.21e+04 $\pm$ 1.35e+04	39.92 $\pm$ 16.95	1.27 $\pm$ 0.3297	1.616 $\pm$ 0.5764
0.3	69.65 $\pm$ 55.94	0.5174 $\pm$ 0.5141	1.171 $\pm$ 0.1988	1.128 $\pm$ 0.494
0.35	19.97 $\pm$ 20.74	0.2676 $\pm$ 0.05661	1.104 $\pm$ 0.07193	0.8891 $\pm$ 0.361
0.4	37.4 $\pm$ 22.78	0.1714 $\pm$ 0.02294	1.445 $\pm$ 0.8694	1.755 $\pm$ 2.934
0.45	10.34 $\pm$ 7.601	0.1757 $\pm$ 0.02442	1.056 $\pm$ 0.08684	1.013 $\pm$ 0.5294
0.5	9.006 $\pm$ 8.21	0.07761 $\pm$ 0.01899	1.147 $\pm$ 0.09186	0.4554 $\pm$ 0.09491
0.55	8.07 $\pm$ 13.2	0.07347 $\pm$ 0.01143	1.059 $\pm$ 0.1069	0.4289 $\pm$ 0.08276
0.6	10.71 $\pm$ 19.87	0.07274 $\pm$ 0.006366	1.157 $\pm$ 0.1726	0.4842 $\pm$ 0.1422
0.65	4.174 $\pm$ 6.635	0.0786 $\pm$ 0.01956	1.077 $\pm$ 0.1054	0.4043 $\pm$ 0.03718
0.7	5.624 $\pm$ 7.853	0.07844 $\pm$ 0.01241	1.238 $\pm$ 0.1772	0.4419 $\pm$ 0.07988
0.75	0.8214 $\pm$ 1.968	0.08304 $\pm$ 0.01412	0.994 $\pm$ 0.1968	0.3944 $\pm$ 0.03835
0.8	2.523 $\pm$ 4.494	0.09615 $\pm$ 0.01516	1.212 $\pm$ 0.2465	0.4615 $\pm$ 0.03094
0.85	1.507 $\pm$ 2.453	0.0243 $\pm$ 0.007353	1.046 $\pm$ 0.3319	0.2773 $\pm$ 0.02827
0.9	2.008 $\pm$ 5.301	0.02633 $\pm$ 0.008995	1.248 $\pm$ 0.3935	0.3223 $\pm$ 0.0247
0.95	2.921 $\pm$ 5.432	0.02504 $\pm$ 0.05607	1.4 $\pm$ 0.6449	0.1362 $\pm$ 0.1281
1.00	0.008248 $\pm$ 0	0.01502 $\pm$ 0	1.045 $\pm$ 0	0.3578 $\pm$ 0

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Table 4: Cross-validation sweep for Mo: held-out energy and force MAE, mean $\pm$ std over 10 stratified draws per fraction (reimplemented quadratic SNAP vs. LRA learned descriptor). Fraction 1.00 is the full-data in-sample training error.

fraction	SNAP δE (eV/atom)	SNAP δF (eV/Å)	LRA δE (eV/atom)	LRA δF (eV/Å)
0.05	1.593 $\pm$ 1.016	4.927 $\pm$ 1.732	219.4 $\pm$ 430.9	39.03 $\pm$ 80.22
0.1	0.8598 $\pm$ 0.5024	4.505 $\pm$ 2.813	12.31 $\pm$ 15.77	4.64 $\pm$ 1.501
0.15	0.2295 $\pm$ 0.07655	1.334 $\pm$ 0.3207	10.07 $\pm$ 11.31	4.232 $\pm$ 2.515
0.2	0.1451 $\pm$ 0.03854	0.891 $\pm$ 0.1857	10.47 $\pm$ 22.77	4.349 $\pm$ 5.664
0.25	0.1181 $\pm$ 0.03388	0.7161 $\pm$ 0.1604	2.444 $\pm$ 0.7565	2.32 $\pm$ 0.4552
0.3	0.09361 $\pm$ 0.01219	0.5364 $\pm$ 0.09484	2.086 $\pm$ 0.465	2.127 $\pm$ 0.2009
0.35	0.08268 $\pm$ 0.01659	0.4263 $\pm$ 0.07231	1.896 $\pm$ 0.2962	1.957 $\pm$ 0.0738
0.4	0.07523 $\pm$ 0.01683	0.455 $\pm$ 0.08976	1.798 $\pm$ 0.3356	1.982 $\pm$ 0.07519
0.45	0.06683 $\pm$ 0.009029	0.3854 $\pm$ 0.1212	1.848 $\pm$ 0.2562	1.968 $\pm$ 0.09961
0.5	0.06418 $\pm$ 0.008535	0.2889 $\pm$ 0.06885	2.094 $\pm$ 0.5768	1.89 $\pm$ 0.09918
0.55	0.06643 $\pm$ 0.00906	0.3082 $\pm$ 0.07753	1.846 $\pm$ 0.3466	1.874 $\pm$ 0.1382
0.6	0.06541 $\pm$ 0.01171	0.3081 $\pm$ 0.09632	1.789 $\pm$ 0.2034	1.901 $\pm$ 0.2263
0.65	0.06683 $\pm$ 0.01223	0.3371 $\pm$ 0.1044	1.829 $\pm$ 0.5017	1.903 $\pm$ 0.2522
0.7	0.06851 $\pm$ 0.01155	0.3234 $\pm$ 0.101	1.706 $\pm$ 0.1667	1.986 $\pm$ 0.2634
0.75	0.07624 $\pm$ 0.02014	0.3426 $\pm$ 0.0981	1.717 $\pm$ 0.2324	1.927 $\pm$ 0.3702
0.8	0.0678 $\pm$ 0.01597	0.3193 $\pm$ 0.1274	1.872 $\pm$ 0.5617	1.819 $\pm$ 0.435
0.85	0.06623 $\pm$ 0.01428	0.3094 $\pm$ 0.1119	1.849 $\pm$ 0.5463	1.86 $\pm$ 0.2949
0.9	0.0633 $\pm$ 0.01041	0.2784 $\pm$ 0.1179	2.041 $\pm$ 1.626	1.893 $\pm$ 0.6243
0.95	0.05308 $\pm$ 0.01443	0.2146 $\pm$ 0.05237	1.424 $\pm$ 0.3272	1.499 $\pm$ 0.6246
1.00	0.05743 $\pm$ 0	0.1556 $\pm$ 0	1.537 $\pm$ 0	1.733 $\pm$ 0

Table 5: Cross-validation sweep for Nb: held-out energy and force MAE, mean $\pm$ std over 10 stratified draws per fraction (reimplemented quadratic SNAP vs. LRA learned descriptor). Fraction 1.00 is the full-data in-sample training error.

fraction	SNAP δE (eV/atom)	SNAP δF (eV/Å)	LRA δE (eV/atom)	LRA δF (eV/Å)
0.05	87.51 $\pm$ 186.2	424.9 $\pm$ 877.3	1.04e+03 $\pm$ 2.74e+03	243.4 $\pm$ 415.8
0.1	0.5842 $\pm$ 0.3586	3.117 $\pm$ 1.694	6.671 $\pm$ 5.991	4.941 $\pm$ 2.22
0.15	0.1767 $\pm$ 0.05927	1.031 $\pm$ 0.2852	5.868 $\pm$ 8.454	3.805 $\pm$ 1.754
0.2	0.0929 $\pm$ 0.03586	0.604 $\pm$ 0.1736	11.05 $\pm$ 23.72	3.978 $\pm$ 2.489
0.25	0.08364 $\pm$ 0.0391	0.5024 $\pm$ 0.1926	3.778 $\pm$ 1.922	3.595 $\pm$ 1.327
0.3	0.08141 $\pm$ 0.03183	0.3945 $\pm$ 0.1673	1.94 $\pm$ 0.5099	2.614 $\pm$ 0.1893
0.35	0.06179 $\pm$ 0.01669	0.2689 $\pm$ 0.05676	1.606 $\pm$ 0.3941	2.678 $\pm$ 0.484
0.4	0.05314 $\pm$ 0.008843	0.2232 $\pm$ 0.03558	1.732 $\pm$ 0.4094	2.711 $\pm$ 0.2579
0.45	0.04705 $\pm$ 0.007206	0.1958 $\pm$ 0.02471	2.987 $\pm$ 0.9358	2.8 $\pm$ 0.1853
0.5	0.04237 $\pm$ 0.002515	0.1668 $\pm$ 0.02693	1.746 $\pm$ 0.5881	2.471 $\pm$ 0.1997
0.55	0.04423 $\pm$ 0.004617	0.1792 $\pm$ 0.03818	2.115 $\pm$ 0.7085	2.827 $\pm$ 0.175
0.6	0.04301 $\pm$ 0.003343	0.1605 $\pm$ 0.03001	2.002 $\pm$ 0.8302	2.621 $\pm$ 0.2341
0.65	0.04331 $\pm$ 0.003165	0.1508 $\pm$ 0.02679	1.89 $\pm$ 0.5472	2.5 $\pm$ 0.2649
0.7	0.04439 $\pm$ 0.003886	0.1522 $\pm$ 0.03292	2.195 $\pm$ 0.7657	2.5 $\pm$ 0.2686
0.75	0.0489 $\pm$ 0.008686	0.1733 $\pm$ 0.05148	2.22 $\pm$ 0.608	2.46 $\pm$ 0.3674
0.8	0.05336 $\pm$ 0.01688	0.2103 $\pm$ 0.07306	2.368 $\pm$ 0.8539	2.87 $\pm$ 0.3015
0.85	0.04539 $\pm$ 0.005274	0.1936 $\pm$ 0.09326	2.135 $\pm$ 0.8468	2.641 $\pm$ 0.5281
0.9	0.05094 $\pm$ 0.01284	0.2056 $\pm$ 0.09157	2.563 $\pm$ 1.384	2.96 $\pm$ 0.6877
0.95	0.0449 $\pm$ 0.01361	0.2136 $\pm$ 0.1889	1.767 $\pm$ 0.6843	3.069 $\pm$ 1.252
1.00	0.04335 $\pm$ 0	0.08611 $\pm$ 0	1.173 $\pm$ 0	2.233 $\pm$ 0

Table 6: Cross-validation sweep for Re: held-out energy and force MAE, mean $\pm$ std over 10 stratified draws per fraction (reimplemented quadratic SNAP vs. LRA learned descriptor). Fraction 1.00 is the full-data in-sample training error.

fraction	SNAP δE (eV/atom)	SNAP δF (eV/Å)	LRA δE (eV/atom)	LRA δF (eV/Å)
0.05	0.4594 $\pm$ 0.3028	3.542 $\pm$ 1.624	2.07e+05 $\pm$ 4.34e+05	9.34e+03 $\pm$ 1.71e+04
0.1	0.1891 $\pm$ 0.05556	2.095 $\pm$ 0.6351	5.48e+04 $\pm$ 1.21e+05	2.19e+03 $\pm$ 4.32e+03
0.15	0.1241 $\pm$ 0.01501	1.243 $\pm$ 0.2362	1.15e+03 $\pm$ 2.72e+03	92.24 $\pm$ 202.1
0.2	0.1222 $\pm$ 0.02607	1.016 $\pm$ 0.2894	94.78 $\pm$ 184.6	25.89 $\pm$ 57.15
0.25	0.1096 $\pm$ 0.02758	0.8145 $\pm$ 0.1364	30.04 $\pm$ 58.53	9.951 $\pm$ 11.4
0.3	0.1103 $\pm$ 0.02186	0.8211 $\pm$ 0.2709	22.22 $\pm$ 60.53	6.654 $\pm$ 10.57
0.35	0.09752 $\pm$ 0.0147	0.6629 $\pm$ 0.09216	6.746 $\pm$ 14.52	3.217 $\pm$ 0.2646
0.4	0.1028 $\pm$ 0.01529	0.6398 $\pm$ 0.1246	7.733 $\pm$ 11.16	3.327 $\pm$ 0.8055
0.45	0.1024 $\pm$ 0.01033	0.5939 $\pm$ 0.09661	1.524 $\pm$ 0.2798	2.992 $\pm$ 0.06619
0.5	0.09892 $\pm$ 0.009244	0.5613 $\pm$ 0.1109	1.395 $\pm$ 0.2063	2.972 $\pm$ 0.1057
0.55	0.09483 $\pm$ 0.008022	0.5014 $\pm$ 0.0571	1.35 $\pm$ 0.11	2.989 $\pm$ 0.1181
0.6	0.09866 $\pm$ 0.01114	0.5522 $\pm$ 0.0491	1.445 $\pm$ 0.2768	2.997 $\pm$ 0.1207
0.65	0.09835 $\pm$ 0.004627	0.5028 $\pm$ 0.04991	1.538 $\pm$ 0.1828	3.043 $\pm$ 0.1445
0.7	0.09238 $\pm$ 0.008358	0.4889 $\pm$ 0.05503	1.709 $\pm$ 0.417	3.025 $\pm$ 0.2252
0.75	0.09269 $\pm$ 0.01411	0.5087 $\pm$ 0.07023	1.644 $\pm$ 0.4948	3 $\pm$ 0.09267
0.8	0.1009 $\pm$ 0.009058	0.5198 $\pm$ 0.1093	1.481 $\pm$ 0.1489	2.899 $\pm$ 0.32
0.85	0.09742 $\pm$ 0.009461	0.5456 $\pm$ 0.1351	1.533 $\pm$ 0.1675	3.101 $\pm$ 0.57
0.9	0.09434 $\pm$ 0.01113	0.4366 $\pm$ 0.04273	1.49 $\pm$ 0.4657	2.666 $\pm$ 0.4716
0.95	0.09687 $\pm$ 0.02274	0.6253 $\pm$ 0.3659	1.523 $\pm$ 0.5487	3.014 $\pm$ 1.023
1.00	0.1024 $\pm$ 0	0.3697 $\pm$ 0	1.28 $\pm$ 0	2.847 $\pm$ 0